

Clinical Risk Assessment Report: Patient 31

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 19.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.90x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 1.9%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Tumor Sphericity (CT): A decrease would alter the prediction outcome.
- Maximum Tumor Density: A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.54 [Top 7%] (+4.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.67 [Avg (67th %ile)] (+2.0%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.52 [Top 5%] (+2.0%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.71 [Top 24%] (-1.9%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.04 [Avg (43th %ile)] (-1.5%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.2 [Avg (51th %ile)] (-1.2%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.49 [Avg (29th %ile)] (-1.1%)**

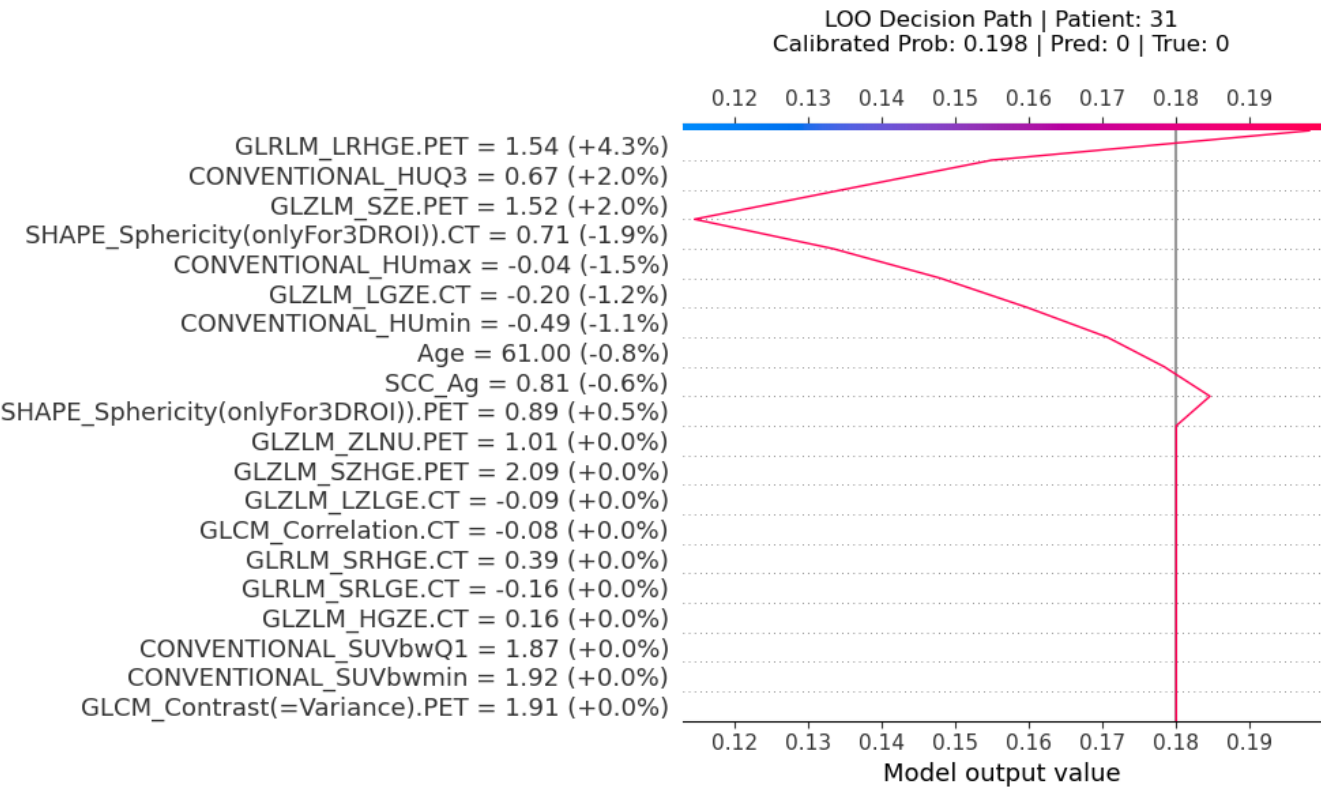
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Patient Age (Age)

Value: 61.0 [Avg (40th %ile)] (-0.8%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

